

Trioron — Results (§4)

4 Results

§4.1 establishes the headline: manifold replay at 30 KB beats fixed-EWC σ -confidently at matched parameters. §4.2 sweeps five continual-learning families against the same protocol. §4.3 traces the storage-vs-accuracy Pareto. §4.4 reports the dream-archive Phase 1 + Phase 2 storage win. §4.5 walks through the ship-wake-extend loop end-to-end at 23 tasks.

4.1 Manifold Replay (Headline)

Table 1 reports the $n = 3$ multi-seed panel on chained-15 for the four primary trioron arms with manifold replay enabled.

Table 1: chained-15, $n = 3$ trioron panel. ★ marks the headline arm.

arm	full	domain	task
<code>fixed_ewc_small</code>	0.585 ± 0.002	0.655 ± 0.006	0.951 ± 0.001
<code>grown_capped_no_dream</code>	0.577 ± 0.011	0.661 ± 0.009	0.951 ± 0.000
<code>grown_capped_dream</code>	0.587 ± 0.008	0.657 ± 0.009	0.956 ± 0.003
<code>grown_uncapped_dream</code>	0.604 ± 0.003	0.679 ± 0.010	0.960 ± 0.001 ★

The headline arm (`grown_uncapped_dream`) reaches 0.604 full / 0.679 domain / 0.960 task at 30 KB of manifold replay storage. Forgetting is negative across all three seeds (-0.566 ± 0.009 on `grown_uncapped_dream`): past-class accuracy *increases* over time after a task ends. The manifold replay block exposes the network to past classes throughout the curriculum, so old-class decision boundaries continue sharpening even after the task is over — qualitatively new behavior relative to any prior rehearsal mechanism, which at best minimize forgetting toward zero.

Paired σ -comparisons (Table 2) confirm that growth + manifold replay σ -confidently beats fixed-EWC at matched trainable parameters.

Table 2: Paired comparisons, $n = 3$ paired by seed.

comparison	metric	mean Δ	σ
<code>fixed_ewc_small</code> vs <code>grown_uncapped_dream</code>	full	-0.018	-6.36
<code>fixed_ewc_small</code> vs <code>grown_uncapped_dream</code>	task	-0.009	-4.08
<code>grown_capped_no_dream</code> vs <code>grown_uncapped_dream</code>	full	-0.027	-3.44
<code>grown_capped_no_dream</code> vs <code>grown_uncapped_dream</code>	domain	-0.018	-18.14

The $+6.4\sigma$ paired margin against `fixed_ewc_small` at full-softmax is the strongest growth-vs-fixed result we have measured in this project. Earlier benches in a regression arena showed

growth-vs-fixed advantages of the same sign but only $+1.0\text{--}1.2\sigma$; manifold replay is the variance-tightening ingredient that lifts a borderline central-tendency win to a σ -robust one.

4.2 Five-Family Competitor Sweep

Table 3 ports the same chained-15 protocol to five major continual-learning families, with the manifold-grown headline included. All results are $n = 3$ multi-seed.

Table 3: chained-15 competitor sweep, $n = 3$. ★ marks family-best on a metric.

family	method	full	domain	task
rehearsal (ours)	manifold-grown (μ, σ)	0.604 ± 0.003	0.679 ± 0.010	0.960 ± 0.001
rehearsal (exemplar)	hippo $K = 50$	0.646 ± 0.009	0.699 ± 0.012	0.966 ± 0.001
partition	PackNet matched	0.074 ± 0.016	0.202 ± 0.028	0.945 ± 0.005
partition	PackNet standard	0.049 ± 0.015	0.140 ± 0.037	0.907 ± 0.006
attention-mask	HAT matched	0.132 ± 0.011	0.232 ± 0.033	0.938 ± 0.006
attention-mask	HAT standard	0.051 ± 0.018	0.147 ± 0.015	0.944 ± 0.009
online-regularization	Online EWC $\gamma = 0.95$	0.187 ± 0.033	0.347 ± 0.027	0.957 ± 0.001
distillation	LwF + EWC	0.199 ± 0.047	0.380 ± 0.052	0.964 ± 0.001 ★

Manifold-grown σ -dominates every named competitor on full-softmax and domain-aware (Table 4).

Table 4: Paired σ -comparisons, manifold-grown vs each competitor.

comparison	full Δ (σ)	domain Δ (σ)	task Δ (σ)
vs PackNet matched	$+0.529$ ($+31\sigma$)	$+0.477$ ($+24\sigma$)	$+0.015$ ($+3.4\sigma$)
vs PackNet standard	$+0.555$ ($+47\sigma$)	$+0.538$ ($+12\sigma$)	$+0.052$ ($+9.8\sigma$)
vs HAT matched	$+0.471$ ($+34\sigma$)	$+0.447$ ($+12\sigma$)	$+0.022$ ($+3.6\sigma$)
vs HAT standard	$+0.553$ ($+27\sigma$)	$+0.531$ ($+47\sigma$)	$+0.016$ ($+1.7\sigma$)
vs Online EWC	$+0.417$ ($+13\sigma$)	$+0.332$ ($+14\sigma$)	$+0.002$ ($+1.6\sigma$)
vs LwF + EWC	$+0.405$ ($+8\sigma$)	$+0.298$ ($+6\sigma$)	-0.005 (-26σ) ★

LwF + EWC marginally beats manifold-grown on task-aware accuracy by 0.005 absolute (-26σ on a tight std). The mechanism is structural: LwF distills the live network’s logits toward the anchored network’s logits on old-class columns, which directly optimizes the within-old-class discrimination that task-aware measures. We name this loss explicitly. On the two harder metrics — full-softmax (no task ID at inference) and domain-aware — manifold-grown wins by $+6$ to $+47\sigma$.

The pattern is consistent with earlier regression-arena results: every borderline central-tendency win in the regression arena (grown vs PackNet, vs HAT, vs Online EWC, vs fixed-EWC) widens to a σ -confident classification-arena win once manifold replay is added. Two interpretations are consistent with the data: (a) manifold replay is the variance-tightening ingredient that lifts central-tendency wins to robust ones; (b) classification with informative Fisher (cross-entropy) is a fairer arena for EWC-family methods than regression with hinge-contrastive (where Fisher saturates near zero) — and yet manifold-grown still dominates.

4.3 Storage-vs-Accuracy Pareto

Manifold replay’s 30 KB sketch is competitive with episodic-exemplar buffers an order of magnitude larger. Table 5 sweeps K (samples per class stored) for the hippocampal-buffer baseline alongside

the manifold sketch. All rows report `grown_uncapped_dream` as the trioron arm.

Table 5: Storage-vs-accuracy Pareto ($n = 3$ means).

variant	storage	full	gap to $K = 50$
engram + diff (no rehearsal)	24 KB	0.260	-0.39
manifold (μ, σ)	30 KB	0.604	-0.04 ★
combined + hippo $K = 10$	174 KB	0.570	-0.08
combined + hippo $K = 20$	324 KB	0.601	-0.05
hippo $K = 50$	750 KB	0.646	—

Manifold Pareto-dominates hippo $K = 10$ (cheaper *and* more accurate), matches hippo $K = 20$ at one-tenth the storage, and lands within 4 absolute points of $K = 50$ at one-twenty-fifth the storage. On task-aware the gap to $K = 50$ is $< 1\%$.

Logit-inversion variants (one-shot DeepInversion, refresh-all, and diversity-regularized) all peaked at 0.27 full — the same engram-attractor pathology in 128-dimensional L0 space. Per-class statistics live on the real L0 manifold by construction; inversion attractors do not. An ablation removing per-class σ (μ -only replay) loses ≈ 0.07 absolute on full-softmax at 30 classes — roughly 12% of manifold replay’s advantage — confirming that per-class noise scale is load-bearing.

4.4 Dream Archive Phase 1 + Phase 2

Phase 1 (mid-curriculum row archive) and Phase 2 (end-of-curriculum int8 quantization of archived rows + BF16 active path) are evaluated as a deployment-side compression layer atop the manifold-grown architecture. Table 6 reports $n = 3$ numbers on the four-arm panel with archive enabled, paired against the no-archive manifold baseline at matched seeds.

Table 6: chained-15 with archive Phase 1 + Phase 2, $n = 3$.

arm	full v2	full Δ vs no-archive	task Δ
<code>fixed_ewc_small</code>	0.572 ± 0.004	-0.013	+0.001
<code>grown_capped_no_dream</code>	0.559 ± 0.004	-0.019	-0.003
<code>grown_capped_dream</code>	0.572 ± 0.009	-0.015	+0.006
<code>grown_uncapped_dream</code>	0.617 ± 0.004	+0.013	+0.006

Phase 1’s training-time effect lies inside seed-to-seed variance: the same code on the same seeds produced `grown_uncapped_dream` $\Delta = +0.013$ in run v2 versus -0.004 in run v1, attributable to torch nondeterminism in archive-trigger timing (task-1 accuracies already differ before any archive fires). We cannot σ -confidently claim Phase 1 helps or hurts during training. Phase 2 (int8 snap of archived rows + BF16 active path) is **lossless within** $\Delta \leq 0.0008$ across every arm and every seed. The BF16 + int8 mixed precision matches FP32 + int8 within $\Delta \leq 0.0008$ as well.

The substantive win is on storage. Table 7 reports BF16-baseline deployment storage with and without the archive + int8 path.

Adding the 30 KB manifold buffer, the headline `grown_capped_dream` deployment image is **157 KB total** (vs 241 KB BF16-baseline + manifold, -35%). At an FP32 deployment baseline the same arm lands at 168 KB FP32 + int8 (-63% relative), but the FP32 framing is the wrong yardstick for a device-conscience deployment; we report BF16 throughout this work because BF16 is the inference dtype.

Table 7: Deployment storage, BF16 baseline, $n = 3$ means.

arm	BF16 baseline	BF16 + int8 archived	Δ
fixed_ewc_small	213.7 KB	130.4 KB	−39%
grown_capped_no_dream	213.7 KB	128.9 KB	−40%
grown_capped_dream	211.2 KB	127.5 KB	−40% ★
grown_uncapped_dream	224.9 KB	137.2 KB	−39%

The -0.015 full-softmax cost on **grown_capped_dream** comes from L1 archive, not L0: by task 15 the archive has typically locked 32–49 of L1’s 48–92 rows, and rows locked mid-curriculum cannot adapt to later tasks. L0 archive is functionally a no-op (L0 is already frozen at init); locking L0 only enables the Phase 2 quantization, where most of the storage saving lives. Future work: more conservative L1 trigger (higher streak threshold, lower *archive_max_per_layer*), or skip L1 archive entirely and rely on L0 quantization alone.

4.5 Extension: Ship-Wake-Extend Loop

We validate the full deployment loop end-to-end on a chained-15 $\rightarrow$ consolidation-dream $\rightarrow$ +8 EMNIST K..R curriculum (1 seed, 4 epochs/task, **grown_capped_dream**).

Table 8: Architecture progression across the ship-wake-extend loop.

phase	layers	trainable	archived	plastic
start	(128, 32, 2)	4 194	[0, 0, 0]	4 194
chained-15 done	(128, 48, 30)	7 662	[104, 32, 0]	3 534
chained-23 done	(128, 80, 46)	14 046	[128, 40, 0]	8 886

The plastic envelope more than doubles during extension (3 534 $\rightarrow$ 8 886 trainable parameters), allocated against the new task budget — but the archived rows from chained-15 are honored throughout. The total stored network grows by only $2.5\times$ while task coverage grows from 30 to 46 classes.

Accuracy at 23 tasks. Full = 0.521, domain = 0.693, task = 0.948. Per-phase task-aware accuracy: MNIST 0.970, Fashion 0.982, EMNIST 0.927 — original tasks survive through the extension at task-aware ≥ 0.93 . Full-softmax dropped from 0.572 (chained-15-only) to 0.521 (chained-23): partly mechanical (46 vs 30 argmax competitors per query, every sample now has more wrong answers to choose from), partly real MNIST drift from extension-time growth (per-phase MNIST full 0.752 $\rightarrow$ 0.678). The -0.014 task-aware drop is small and is the right metric for trioron’s deployment role of context-gated personalization atop a frozen LLM.

Deployment storage (BF16 baseline). 121.3 KB network (L0 = 98.8, L1 = 15.3, head = 7.3) plus ≈ 47 KB manifold buffer (46 classes $\times$ 256 floats) ≈ 168 **KB total** — +11 KB over the chained-15 deployment for 8 additional task capabilities.

The verdict: a 157 KB chained-15 model can be shipped, woken on K..R data in the field, allowed to grow plastic substrate (16 $\rightarrow$ 40 plastic L1 rows), and re-shipped at 168 KB with original tasks preserved at task-aware ≥ 0.97 . The architecture supports continual learning *across deployment cycles*, not only across task boundaries within a single training run.

Limits we are honest about. Full-softmax on this cohort is data-bounded. Per-dataset MLP joint-training upper bounds are MNIST $\approx 98\%$, Fashion-MNIST $\approx 92\text{--}93\%$, EMNIST-letters $\approx 91\text{--}92\%$; the combined 46-class weighted ceiling is ≈ 0.927 . Genuine class confusions (T-shirt / shirt / pullover; I / L; O / Q; B / 8) make 99.9% full-softmax unreachable on this benchmark at any parameter count under an MLP head, and we do not promise it. The defensible claim is: **0.95+ task-aware on 23 class-incremental tasks at 168 KB total deployment, with on-device continual learning capability across deployment cycles** — a regime no existing method occupies.

4.6 Multi-Branch Absorption

Extension within a single organism (§4.5) shows the architecture supports continual learning across deployment cycles. Multi-branch absorption asks the next question: can two organisms trained *independently* on disjoint task streams be *composed* into a single recipient with no retraining of either? With the shared L0 invariant (§??) this is paste-and-go.

Setup. Donors are produced by separate runs of the same `grown_uncapped_dream` arm at L0 seed 42 on disjoint chained-15 / chained-extension sub-blocks: `digits` (MNIST, classes 0–9), `fashion` (Fashion-MNIST, 10–19), `emnist` (EMNIST *A–J*, 20–29), `emnist_kt` (EMNIST *K–T*, 30–39), `emnist_uz` (EMNIST *U–Z* partial, 40–45). Each donor reaches its own task-aware ceiling. We assemble a recipient organism by loading branches in increasing- N prefixes, evaluating on the union test set under *soft + per-branch log-softmax* routing.

Headline: lossless task-aware out to $N = 5$. Donor-standalone task-aware accuracy (averaged over the donors in the prefix) is the upper bound; the assembled organism matches it exactly at every N tested.

N	upper bound (TA)	organism (TA)	gap	full-union
1	0.9809	0.9809	+0.0000	0.6910
2	0.9823	0.9823	+0.0000	0.6099
3	0.9695	0.9695	+0.0000	0.5314
4	0.9624	0.9624	+0.0000	0.4716
5	0.9616	0.9618	+0.0002	0.4462

The bleed-flip point we were hunting — where soft-routing bleed flips from additive to destructive as branches multiply — did not appear at $N = 5/46$ classes on chained-15-scale data. The actual limit at this scale is dataset variety; we ran out of disjoint class blocks within $\text{MNIST} \cup \text{Fashion-MNIST} \cup \text{EMNIST}$.

Bleed manifests where you’d predict, but does not damage task-aware. Diagnostic mean per-branch gate weight per task at $N = 5$:

task (true branch)	digits	fashion	emnist	emnist_kt	emnist_uz
<code>mnist_0_1</code> (digits)	0.81	0.00	0.03	0.14	0.02
<code>mnist_4_5</code> (digits)	0.74	0.03	0.05	0.16	0.03
<code>fashion_0_1</code> (fashion)	0.01	0.97	0.01	0.01	0.00
<code>emnist_8_9</code> (emnist)	0.49*	0.02	0.48*	0.01	0.00

Digit inputs pull up to 0.16 gate weight on `emnist_kt` (O resembles 0, S resembles 5 in code space). EMNIST letters I, J yield a near 50/50 digits/emnist gate split (starred). Yet task-aware accuracy on `emnist_8_9` stays at 0.9244, identical to donor-standalone, because per-branch log-softmax pushes non-covered slot contributions to $-\infty$ in log-probability space — the digits branch’s 0.49 gate cannot win the within-task argmax over class slots 28, 29 that only the `emnist` branch covers. **Soft-routing bleed is benign under principled composition.**

Comparison vs Branch-Train-Merge family ($N = 3$). We benchmark against two BTM variants on the same donors, replicating both the parameter-averaging and learned-router lines of the BTM / Branch-Train-Mix family (Li et al., 2022; Sukhbaatar et al., 2024).

method	zero-shot	task-aware	full-union	training cost
donor-standalone (upper bound)	—	0.9695	—	—
Ours (archive route, soft+norm)	✓	0.9695	0.5314	0
BTM-avg (param averaging)	✓	0.7396	0.0845	0
BTM-MoE (learned router, soft+norm)	✗	0.9695	0.5610	400 steps
BTM-MoE (learned router, raw)	✗	0.9695	0.1725	400 steps

BTM-avg averages L1 substrate across donors entry-wise; the resulting hidden units are the mean of three different feature detectors and none of the donors’ specialty features survive. This replicates the documented BTM-avg weakness (-0.23 absolute task-aware here). BTM-MoE substitutes our archive-likelihood gate with a learned MLP router trained 400 steps on $(L0(x), \text{branch_id})$ pairs from each donor’s training images; it ties our task-aware exactly at the upper bound but requires the router-training pass on real data. The zero-shot AND lossless intersection is what this architecture occupies.

Dream-cycle calibration as a tunable knob. For deployments where full-softmax is load-bearing, a tiny additive bias offset trained on synthetic samples drawn from each branch’s archive (no real data, no L1 retraining) closes most of the calibration gap.

variant	params	Δ task-aware	Δ full-union
no calibration (baseline)	0	—	—
per-branch bias (lr= $5e-2$, 400 st)	3	+0.0000	+0.0018
per-class bias (lr= $5e-3$, 200 st)	30	-0.0005	+0.0047
per-class bias (lr= $5e-2$, 400 st)	30	-0.0021	+0.0245

The per-branch knob is mathematically task-aware-safe (one uniform per-donor shift is monotonic on within-task argmax under disjoint coverage). The per-class knob trades a small amount of task-aware for a larger full-softmax lift, landing at full-union 0.5560 — $\approx 80\%$ of the BTM-MoE level (0.5610) without the router-training pass on real data. Storage cost: 30 floats $\times$ 4 B = 120 B at $N = 3$, $< 0.03\%$ of the organism.

Storage at $N = 3$. Shared $L0 = 401$ KB (fixed regardless of N); branch substrate ≈ 90 KB; archives ≈ 30 KB. **Per-skill marginal cost is ≈ 40 KB** (substrate + archive); $L0$ amortizes across all branches. A device that ships with the $L0$ footprint already present can absorb new skills at 40 KB each.

The verdict. The architecture supports zero-shot lossless composition of independently-trained organisms under the shared-L0 invariant. We tie the strongest BTM baseline on the task-aware metric while requiring zero training; we beat the parameter-averaging baseline by 0.23 absolute. The mechanism is robust to soft-routing bleed at $N \leq 5$ on chained-15-scale data, with per-branch log-softmax as the load-bearing inference-only calibration step.

Limits we are honest about. This is $N \leq 5$ on a 46-class image-classification benchmark. LLM-scale Branch-Train-Merge benchmarks — where BTM-MoE was originally evaluated — have qualitatively different dynamics in expert specialization, gate-network capacity, and class-space size; we make no scale-extrapolation claim. The shared-L0 invariant is a hard prerequisite: organisms with mismatched L0 seeds require a distillation-through-paired-codes step that we have not measured here and which is no longer instantaneous. Saturation must exist at some N ; we did not find it within the data variety chained-15 supports.

References

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